The FLAIR Privacy Boundary

Code crosses rightward · only aggregate JSONs return · patient data does not.

PUBLIC SIDE · researcher · Model Owner · FLAIR Portal · leaderboard

BARRIER

12+ INSTITUTIONAL SILOS · data stays here · no internet · no root · immutable image

DEVELOP — train against MIMIC-CLIF

Researcher

MIMIC-CLIF
~70k stays · public

FLAIR CLI

`flair load-task` → cohort + splits
train model → y_true · y_pred · y_prob
`flair build-report` → 4 aggregate JSONs

TRIPOD+AI-compliant outputs

Container
model + code · frozen

Model Owner submits container

code in

containerised code

same bytes execute at e

PRIVACY BOUNDARY

patient data never crosses

`table1 tripod fairnesscurves`
4 JSONs

aggregate JSONs only · cell-suppress

aggregates out

Site 1	Site 2	Site 3	Site 4
CLIF tables · local	CLIF tables · local	CLIF tables · local	CLIF tables · local
Site Lead	Site Lead	Site Lead	Site Lead
HIPAA	HIPAA	HIPAA	HIPAA

Site 5	Site 6	Site 7	Site 8
CLIF tables · local	CLIF tables · local	CLIF tables · local	CLIF tables · local
Site Lead	Site Lead	Site Lead	Site Lead
HIPAA	HIPAA	HIPAA	HIPAA

Site 9	Site 10	Site 11	Site 12+
CLIF tables · local	CLIF tables · local	CLIF tables · local	CLIF tables · local
Site Lead	Site Lead	Site Lead	Site Lead
HIPAA	HIPAA	HIPAA	HIPAA

each silo: container runs in CLIF-WorkBench; only aggregate JSONs leave the site

GOVERN — Benchmark Committee

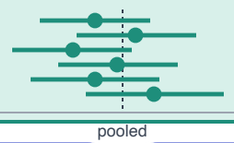
reviews task PRs (single-file modules) · approves models
clinicians · methodologists · regulators · patient reps

approval gates what crosses the boundary & runs at sites

FLAIR PORTAL — orchestrate · register · verify

FLAIR Portal
registers tasks + models · distributes containers · hosts models · verifies SHA-256 · merges

PUBLISH — public leaderboard



Per-site AUROC
forest plot
95% bootstrap CI
variance is the signal

pooled
0.84
[0.79–0.88]

T1 · ICU mortality | **T2 · LTACH discharge** | **T3 · Extubation failure** | **T4 · Sepsis (Abx-anchored)** | **T5 · AKI (KDIGO-2)** | **+ by community PR**

HIPAA-safe
cell-suppressed at n<10

catalog grows by GitHub pull request · acceptance via MIMIC-CLIF reproducibility + clinical review + site-governance review